

BRUCE DAVIS

AI SYSTEMS ENGINEER · AGENTIC AI · IAM · SECURITY

TARGET ROLE: AI Systems Engineer · Forward Deployed AI Engineer · Applied AI Engineer · Senior AI Engineer – Agentic Systems
Open to full-time, contract, and project work. Remote-first.

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Portfolio

I build agentic AI systems — MCP servers, ReAct agent loops, multi-agent orchestration, fine-tuned classifiers, Microsoft Copilot deployments, RAG pipelines. The fifteen years before that were in identity and security: **Non-Human Identity (NHI) governance at Microsoft**, HIPAA-compliant IAM for 16,000+ users at Veradigm, payment infrastructure for SWIFT/CHIPS/Fedwire at Wells Fargo. Deploying an AI agent safely is mostly an identity and authorization problem, which is the work I already know. PMP-certified.

SELECTED PROJECTS – AGENTIC AI, ML, IDENTITY

AUTONOMOUS SECURITY TRIAGE AGENT

ReAct agent loop with human-in-the-loop authorization on destructive actions

ReAct agent on Gemini 2.5 Flash — reads logs, enriches threat intel, proposes containment through function-calling tools. Destructive actions pause for explicit human approval. An autonomy dial scopes operational reach. Built for security teams that need AI assistance without surrendering control. Python · Streamlit · Gemini 2.5 Flash · google-genai SDK · ReAct · Function Calling · HITL.

AI AGENT GOVERNANCE CONSOLE

Non-Human Identity governance applied directly to a fleet of AI agents

Six AI agents as discoverable, classified, owned machine identities — scoped credentials, intake-and-approval workflow, rotation, decommissioning in one trail. Generative UI on a posture event stream. The agent fleet itself becomes an NHI governance surface. Python · React · TypeScript · Multi-agent Architecture · NHI · OAuth · Generative UI.

NHI CREDENTIAL RISK CLASSIFIER

Fine-tuned DistilBERT classifier — 99.33% accuracy, 10.8ms / record on CPU

Synthetic dataset, two fine-tuned models (DistilBERT for production triage, Qwen 2.5 1.5B + LoRA for cachet), full eval harness with per-class P/R/F1, confusion matrix, latency, and cost. Becomes the brain of the Governance Console's Triage agent. Hugging Face Transformers · PEFT / LoRA · PyTorch · scikit-learn · FastAPI · Python.

ENTRA ID NHI AUDIT

PowerShell module that audits an Entra ID tenant for non-human identity risk

Four cmdlets (Connect, Get-Inventory, Test-Risk, Export-Report) score every service principal, app registration, and managed identity on eight risk factors. Mock fixtures so it runs offline; **GRAPH_MODE=Live** swaps in real Microsoft Graph via managed identity. Self-contained HTML report + structured JSON for SIEM ingest. PowerShell 7 · Microsoft Graph SDK · Entra ID · Active Directory · Bicep-ready.

IT SUPPORT COPILOT FOR TEAMS

Teams-deployed Microsoft Copilot — LLM understands, deterministic policy decides

Bot Framework v4 on the Teams side, Azure OpenAI for reasoning, Microsoft Graph for identity. Four tools and four outcomes (self_serve / auto_fulfill / escalate / deny). LLM picks tools; a separate rule-based policy engine decides what happens — auditable and prompt-injection-resistant. Container Apps + Bicep for production. Bot Framework · Microsoft Graph · Azure OpenAI · Entra ID · FastAPI · Bicep · Python.

WORKFLOW-DRIVEN TRIAGE AGENT

Three protocols, one stack — Google ADK + MCP + n8n

n8n receives a support-ticket webhook, calls a Google ADK agent (Gemini 2.5 Flash) over HTTP, agent calls three MCP tools (intent, customer history, KB) over stdio, n8n routes by priority. Each protocol at the layer it's best for. docker-compose stack; no-n8n demo path for fast iteration. Google ADK · MCP / FastMCP · n8n · Gemini 2.5 · FastAPI · Docker Compose.

LINEAR MCP SERVER

Production MCP server with documented threat model and 30-case eval harness

Six opinionated MCP tools for engineering team operations — inbox triage, issue scoping, velocity audit, status reporting. 30-case eval harness, documented threat model, per-tool cost analysis. Read-only by design — destructive operations never written. Open source. TypeScript · @modelcontextprotocol/sdk · Linear GraphQL API · Zod · Vitest.

WORK

Business Execution Consultant — Solution Design

Nov 2024–Apr 2025

Wells Fargo · Contract

- Mapped network dependencies and ran gap analysis across SWIFT, CHIPS, and Fedwire payment infrastructure — documented system flows that hadn't existed on paper, accelerated regulatory approval, and cut design rework by 25%.
- Designed NIST-compliant documentation architecture for network security controls; became the cross-team reference layer for incident response.
- Decomposed executive payment-rail objectives into detailed execution plans for ISO 20022 interoperability and Federal Reserve compliance.

Microsoft · Contract

Forward-deployed Cloud Solutions Architect aligned to Microsoft's AI Factory — embedded with enterprise customers to design, ship, and deploy Azure AI solutions.

- Built Python automation on the Microsoft Graph API for Entra ID Governance — extracted and normalized identity data into executive-ready dashboards and architecture documentation.
- Architected and shipped a Generative AI workflow MVP on Microsoft Copilot — accelerated knowledge transfer and standardized requirements gathering across global delivery teams.
- Translated Non-Human Identity (NHI) governance capabilities into customer-facing solution designs.

Senior Security Engineer

Apr–Sept 2023

Starbucks · Contract

- Engineered PowerShell automation across 60+ domains for identity lifecycle and access control remediation — shipped scripts and runbooks at the engineering layer.
- Closed 50+ vulnerabilities per sprint through automation and scripted remediation rather than manual triage.
- Hardened identity and access controls across distributed teams — configured authentication protocols, owned security testing, and ran defect remediation end to end.

Project Manager — Privileged Access Management (PAM)

Jul 2021–Nov 2022

Bank of America · Contract

- Led zero-downtime IAM and infrastructure modernization for 12+ teams and 60+ engineers across Oracle, CyberArk, Azure, Middleware, and API platforms — coordinated dependency mapping, change validation, and stakeholder alignment.
- Directed HashiCorp Vault enterprise rollout and service account password rotation automation — integration planning, stakeholder communication, and change management.
- Owned ServiceNow dashboards as operational source of truth for change approvals, release readiness, and migration tracking; ensured all initiatives met stringent audit requirements.
- Authored runbooks, migration plans, and governance documentation for senior leadership; identified delivery risks through deep-dive requirements workshops.

Senior IAM Security Engineer

Jan 2019–Jun 2021

Veradigm · Healthcare

- Architected and shipped an IAM platform for 16,000+ users in a HIPAA-regulated healthcare environment — Entra ID, MIM, and CyberArk integrated end to end.
- Built Power Platform dashboards that surfaced compliance state across the organization; moved audit compliance from 65% to 95% over the engagement.
- Designed and executed a multi-year IAM strategy in a regulated healthcare environment, overseeing technology selection and solution architecture.

Earlier: Data Quality Analyst · Wells Fargo (2017–2018) · IAM Consultant · Cognizant (2016–2017) · Support Escalation Engineer · Microsoft (2012–2015)

STACK

FOUNDATION & SMALL MODELS: OpenAI · Anthropic Claude · Google Gemini · Meta Llama · Qwen 2.5 · Mistral · DistilBERT · Hugging Face · Open-source models

ML & FINE-TUNING: PyTorch · Hugging Face Transformers · PEFT / LoRA · scikit-learn · Eval Harness Design · Synthetic Dataset Generation · Per-Class Metrics & Confusion Matrices

ORCHESTRATION & FRAMEWORKS: MCP · FastMCP · Google ADK · n8n · Microsoft Bot Framework · LangChain · LlamaIndex · Vertex AI · Azure OpenAI · ReAct · Function Calling · Multi-Agent Architecture · RAG · HITL · Anthropic SDK · OpenAI SDK · google-genai SDK · Vercel AI SDK

APPLICATION & UI: Microsoft Copilot · Microsoft Teams · Claude · ChatGPT · Gemini · Notion AI

CLOUD & INFRASTRUCTURE: Microsoft Azure · Azure OpenAI · Azure Container Apps · Azure Bot Service · Application Insights · Bicep · Google Cloud Platform · Vercel · Docker · Docker Compose

IDENTITY & SECURITY: Non-Human Identity (NHI) · Entra ID Governance · Microsoft Graph SDK · CyberArk · Microsoft Identity Manager (MIM) · HashiCorp Vault · Active Directory · Zero Trust · HIPAA · NIST · ISO 20022 · SWIFT · CHIPS · Fedwire · Threat Modeling · ServiceNow

ENGINEERING: Python · TypeScript · JavaScript · Node.js · PowerShell 7 · React · Next.js · FastAPI · Streamlit · Pydantic · JSON · JSON Schema · SQL · GraphQL · Zod · Vitest · pytest · Git · Tailwind

DELIVERY: PMP · Scrum (PSM I) · Microsoft AI Factory · Forward Deployed Engineering · Audit-Ready Documentation

CERTIFICATIONS & EDUCATION

Google UX Design Professional Certificate

Google (via Coursera) · 2023

Certified in Cybersecurity (CC)

ISC2

Project Management Professional (PMP)

Project Management Institute

M.S., Computer Graphics Technology

Purdue University

Thesis: An Exploration of the Role of Interactivity in Video Game Based-Learning

B.S., Computer Science

Lincoln University

